

Notes: Baghai, Servaes & Tamayo (JF, 2014)

Have Rating Agencies Become More Conservative? Implications for Capital Structure and Debt Pricing

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1 Big picture

- **Question.** Have corporate rating agencies become more conservative over time, and does that change corporate financing and bond pricing?
- **Main fact.** From 1985 to 2009, holding firm characteristics fixed, average corporate ratings fall by about three notches.
- **Not the structured-finance story.** The paper studies corporate issuer ratings; unlike the common post-crisis narrative about inflated structured-product ratings, corporate ratings become harsher.
- **Validation.** If stricter ratings are warranted by unobserved risk, default rates within ratings should not fall. But default rates fall for many categories.
- **Corporate effects.** Firms hit harder by conservatism issue less debt, have lower leverage, hold more cash, are less likely to obtain public bond ratings, and grow less.
- **Bond-market effects.** Holding actual rating fixed, firms rated worse than the old model predicts have lower spreads. Markets partly undo the conservative rating, but not fully.
- **Takeaway.** Credit ratings are not just labels. Shifts in rating standards affect capital structure, market access, cash policy, growth, and debt pricing.

2 Incremental contribution of the paper

1. **Long corporate-rating trend.** Extends Blume, Lim, and MacKinlay-type evidence on tighter corporate rating standards through 2009.
2. **Consequences, not only documentation.** Links rating conservatism to debt issuance, leverage, cash, public bond access, growth, investment, acquisitions, and spreads.
3. **Firm-level conservatism measure.** Defines Rat_Diff as actual rating minus an old-standard predicted rating, making rating standards usable in firm-level regressions.
4. **Default-rate validation.** Tests whether tighter ratings are warranted using default rates within rating categories.
5. **Debt-pricing discipline.** Shows bond investors price old-model predicted ratings even after actual ratings are controlled for.
6. **Capital-structure implication.** Provides a ratings-based explanation for underleverage and constrained public debt access.

3 Data and measurement

- **Ratings.** S&P long-term domestic issuer ratings from Compustat, 1985-2009. Ratings are coded numerically: $AAA = 1, AA+ = 2, \dots, C = 21$, so larger numbers mean worse ratings.
- **Sample restrictions.** Exclude financials, utilities, governments, quasi-governmental entities, missing key Compustat data, and affiliates.
- **Ratings controls.** Leverage, cash, debt/EBITDA, interest coverage, profitability, volatility, size, tangibility, CAPEX, beta, idiosyncratic risk, industry and/or firm fixed effects.
- **Conservatism measure.** Estimate ratings models in 1985-1996, predict ratings in 1997-2009, and compute

$$Rat_Diff_{it} = ActualRating_{it} - PredictedRating_{it}^{1985-1996}.$$

Positive Rat_Diff means the firm is rated worse than it would have been under old standards.

- **Validation and outcomes.** Moody's five-year default rates, recovery rates, Compustat capital-structure variables, bond rating access, cash/assets, growth, and Merrill Lynch bond spreads.

4 Empirical specifications and identification design

4.1 Ratings trend

$$Rating_{it} = \alpha + \beta' X_{it} + \sum_t \gamma_t Year_t + \varepsilon_{it}.$$

Interpretation: A positive and rising γ_t means that the same observable firm receives a worse rating later in time. Firm fixed effects compare a firm with itself across years.

4.2 Default-rate validation

$$DefaultRate_{r,t} = \alpha_r + \delta Trend_t + \theta Recession_t + u_{r,t}.$$

Interpretation: Falling defaults within rating categories support the view that ratings have become stricter rather than merely tracking hidden risk.

4.3 Corporate policy effects

$$Y_{it} = \alpha + \beta Rat_Diff_{i,t-k} + \lambda Rating_{i,t-k} + \Gamma' Controls_{i,t-k} + FE + \varepsilon_{it}.$$

Interpretation: Actual ratings are controlled for, so β captures the extra effect of being rated worse than the old-standard model predicts.

4.4 Spread regressions

$$Spread_{j,i,t} = \beta_1 ActualRating_{j,i,t} + \beta_2 Rat_Diff_Firm_{i,t} + \beta_3' Controls_{j,i,t} + FE + \varepsilon_{j,i,t}.$$

Interpretation: If $\beta_2 < 0$, investors charge lower spreads to firms whose actual ratings look excessively harsh, conditional on actual rating.

4.5 Identification logic

- The core evidence is a controlled shift in the rating-characteristics mapping.
- Default and recovery rates test whether the shift is warranted by risk.
- Old-model predicted ratings translate aggregate standards into firm-level treatment intensity.
- Capital-structure, access, and cash regressions test real corporate consequences.
- Spread regressions test whether investors agree that actual ratings are too harsh.

Interpretation: The identification is not a randomized shock. It is strongest because four pieces align: ratings become harsher, default rates decline within rating categories, firms adjust financial policies, and markets partly reverse conservative ratings in spreads.

5 Tables and figures

5.1 Table I: Number of Companies by Year and S&P Rating Category; Table II: Summary Statistics

Read: Table I shows the rating distribution by year. The rated sample grows strongly in the 1990s and shifts toward BBB/BB/B categories. Table II shows that rating levels and firm characteristics change over time, which motivates controls.

Economic message: Raw rating deterioration is visible, but the paper's key question is whether it remains after controlling for fundamentals.

Table I
Number of Companies by Year and S&P Rating Category

This table contains the distribution of ratings for our sample firms over time. The ratings have been obtained from the Compustat Ratings File.

Year	Rating									Total
	AAA	AA	A	BBB	BB	B	CCC	CC	C	
1985	26	98	166	104	100	129	5	1	0	629
1986	30	110	201	150	165	253	42	0	0	951
1987	32	111	194	142	183	284	44	1	0	991
1988	35	88	208	146	167	293	32	0	0	969
1989	36	84	199	149	166	273	33	0	1	941
1990	34	86	194	158	151	206	38	3	1	871
1991	33	87	200	163	158	171	34	10	0	856
1992	32	85	200	193	183	170	27	8	0	898
1993	29	84	204	207	232	195	14	1	0	966
1994	28	83	200	227	254	216	17	0	1	1,026
1995	30	75	225	241	269	234	18	0	0	1,092
1996	28	86	227	279	296	267	16	3	0	1,202
1997	26	84	232	319	334	323	12	2	0	1,332
1998	25	82	240	351	379	368	30	7	0	1,482
1999	18	69	216	374	385	421	36	9	0	1,528
2000	14	51	237	378	374	422	45	7	0	1,528
2001	13	47	221	389	375	375	61	12	0	1,493
2002	11	41	214	381	390	337	72	16	0	1,462
2003	11	38	201	376	407	382	60	6	0	1,481
2004	9	37	198	371	431	376	42	4	0	1,468
2005	9	34	197	355	416	363	46	2	0	1,422
2006	9	34	170	342	391	393	36	2	0	1,377
2007	7	32	162	321	363	359	25	4	0	1,273
2008	7	31	152	314	324	328	50	12	0	1,218
2009	5	32	147	316	303	330	43	4	0	1,180
Total	537	1,689	5,005	6,746	7,196	7,468	878	114	3	29,636

Table I. Rating counts by year and category; raw ratings migrate toward worse categories.

Table II
Summary Statistics: Ratings Regressions

This table presents annual averages of the variables employed in the ratings regressions. *Rating*, the dependent variable, is the numerical equivalent of the rating where AAA is 1, AA+ is 2, AA is 3, etc. *IntCov* is EBITDA/Interest Expense. *Profit* is EBITDA divided by sales. *Book Lev* is long- and short-term debt divided by total assets. *Size* is the log of the book value of assets in constant 2005 dollars. *Debt/EBITDA* is long- and short-term debt divided by EBITDA. *Neg. Debt/EBITDA* is an indicator variable set equal to one if EBITDA is negative and zero otherwise. *Vol* is the volatility of *Debt/EBITDA*. *CapEx/Assets* is capital expenditures divided by total assets. *PPE/Assets* is net property, plant, and equipment divided by total assets. *CAPEX/Assets* is capital expenditures divided by total assets. When the ratio of long-term debt and short-term debt to EBITDA is negative, we set it equal to zero. For firms with zero interest payments we set *IntCov* equal to the 99th percentile of its distribution. All explanatory variables except for *Size* and *Neg. Debt/EBITDA* are winsorized at the 99th percentile; profitability, interest coverage, and the volatility of profitability are also winsorized at the 1st percentile.

Year	Rating	IntCov	Profit	Book Lev	Size	Debt/EBITDA	Neg. Debt/EBITDA	Vol	Cash/Assets	ConsDebt/Assets	Rest/Assets	PPE/Assets	CAPEX/Assets
1985	8.658	7.328	0.172	0.396	7.469	3.611	0.034	0.036	0.077	0.043	0.018	0.475	0.987
1986	9.830	6.865	0.151	0.366	7.177	3.825	0.057	0.049	0.088	0.057	0.020	0.422	0.880
1987	10.085	6.594	0.165	0.387	7.224	3.773	0.045	0.046	0.088	0.062	0.020	0.406	0.775
1988	10.029	6.124	0.169	0.400	7.387	4.094	0.031	0.045	0.075	0.053	0.021	0.406	0.977
1989	10.020	5.884	0.165	0.417	7.500	3.827	0.039	0.047	0.069	0.047	0.021	0.413	0.075
1990	9.742	5.735	0.157	0.411	7.629	3.792	0.030	0.040	0.064	0.042	0.021	0.420	0.076
1991	9.586	5.690	0.162	0.399	7.649	3.955	0.030	0.036	0.064	0.041	0.022	0.430	0.070
1992	9.561	6.769	0.160	0.389	7.632	3.893	0.027	0.033	0.064	0.043	0.023	0.435	0.069
1993	9.637	7.607	0.165	0.376	7.621	3.696	0.025	0.037	0.070	0.045	0.022	0.433	0.072
1994	9.844	8.144	0.169	0.381	7.605	3.675	0.031	0.041	0.064	0.040	0.021	0.425	0.075
1995	9.964	7.998	0.177	0.386	7.624	3.429	0.030	0.046	0.066	0.033	0.021	0.414	0.078

(Continued)

Table II, part 1. Early sample summary statistics used in ratings regressions.

Table III
Ratings Models

Panel A reports the coefficients for regression models of credit ratings. Models (1) and (3) are OLS regressions with industry dummies based on three-digit SIC codes; models (2) and (4) are ordered logit regressions with industry dummies; and models (5) to (8) are OLS regressions with firm dummies. *Rating* is the stock's DuPont-adjusted beta (one lead and one lag term) computed with daily returns. *Idio*, Risk is the root mean squared error from a regression of daily stock returns on CRSP value weighted index returns. These two variables are standardized by dividing them by their annual cross-sectional means. *Rate* and *Idio*, Risk are winsorized at their 1st and 99th percentiles prior to standardization. *Inf*, Rate is the inflation rate (in %) from the Bureau of Labor Statistics Consumer Price Index (All Urban Consumers); *GDP*, GDP growth is the real GDP growth (in %) from the Bureau of Economic Analysis; *TS Slope* is the slope of the term structure (in %), computed as the yield on the constant maturity 10-year Treasury bond minus the yield on the constant-maturity three-month T-bill; *P/E* is the aggregate price-to-earnings ratio based on the previous year's earnings (from Shiller); and *VIX* is the annual average of the market volatility index, from CBOE (available from 1986). *Linear Trend* takes the value of 0 in 1986, 1 in 1987, 2 in 1988, 3 in 1989, etc. All other variables are described in Table II. The sample in columns (1) to (6) is from 1985 to 2009, while in columns (7) and (8) is from 1986 to 2009. Standard errors are clustered at the firm level in columns (1) to (6) and clustered at the year level in columns (7) and (8), and are robust to heteroskedasticity and autocorrelation. *p*-values are reported in parentheses next to the coefficients. Panel B reports the marginal effects for the ordered logit regression in specification (4) of Panel A. We compute the probability that a firm obtains various ratings in 1985 and in 2009, setting all its characteristics equal to the sample mean.

	Panel A: Ratings Regressions							
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	OLS	Ord. Logit	OLS	Ord. Logit	OLS	OLS	OLS	OLS
IntCov	-0.034 (-0.01)	-0.024 (-0.01)	-0.032 (-0.01)	-0.026 (-0.01)	-0.017 (-0.01)	-0.017 (-0.01)	-0.016 (-0.01)	-0.017 (-0.01)
Profit	-1.620 (-0.01)	-1.465 (-0.01)	-0.635 (0.07)	-0.962 (-0.01)	-0.617 (0.05)	-0.672 (0.05)	-0.776 (0.01)	-0.735 (0.01)
Book Lev	3.648 (-0.01)	3.409 (-0.01)	2.338 (-0.01)	2.706 (-0.01)	2.965 (-0.01)	2.685 (-0.01)	2.565 (-0.01)	2.531 (-0.01)
Size	-1.246 (-0.01)	-1.058 (-0.01)	-1.144 (-0.01)	-1.055 (-0.01)	-1.028 (-0.01)	-0.976 (-0.01)	-0.949 (-0.01)	-0.958 (-0.01)
Debt/EBITDA	0.131 (-0.01)	0.132 (-0.01)	0.090 (-0.01)	0.105 (-0.01)	0.058 (-0.01)	0.040 (-0.01)	0.039 (-0.01)	0.041 (-0.01)
Neg. Debt/EBITDA	1.255 (-0.01)	1.473 (-0.01)	0.617 (-0.01)	0.983 (-0.01)	0.901 (-0.01)	0.469 (-0.01)	0.450 (-0.01)	0.465 (-0.01)
Vol	2.385 (-0.01)	2.597 (-0.01)	1.392 (-0.01)	1.920 (-0.01)	0.803 (0.05)	0.635 (0.14)	0.569 (0.04)	0.672 (0.01)
Cash/Assets	0.064 (0.06)	0.556 (0.06)	0.189 (0.50)	0.307 (0.32)	-0.196 (0.50)	-0.028 (0.50)	0.242 (0.30)	0.177 (0.30)
ConsDebt/Assets	1.778 (-0.01)	1.296 (-0.01)	1.648 (-0.01)	1.619 (-0.01)	0.672 (0.06)	0.505 (0.13)	0.650 (-0.01)	0.559 (-0.01)
Rest/Assets	5.131 (-0.01)	5.666 (-0.01)	4.330 (-0.01)	5.278 (-0.01)	2.668 (0.18)	1.832 (0.34)	1.596 (0.08)	1.296 (0.15)
PPE/Assets	-1.718 (-0.01)	-1.268 (-0.01)	-0.550 (0.36)	-0.420 (0.69)	-0.768 (0.50)	-0.944 (0.51)	-1.037 (-0.01)	-0.980 (-0.01)
CAPEX/Assets	-2.538 (-0.01)	-2.816 (-0.01)	-3.189 (-0.01)	-3.456 (-0.01)	-4.804 (-0.01)	-3.910 (-0.01)	-3.960 (-0.01)	-4.042 (-0.01)
Rate			0.382 (-0.01)	0.243 (-0.01)			0.079 (-0.01)	0.084 (0.01)

(Continued)

Table II, part 2. Mid-sample summary statistics.

Table II, part 3. Late-sample summary statistics and notes.

5.2 Table III: Ratings Regressions; Figure 1: Plot of Year Dummies

Read: Table III estimates the ratings model; Figure 1 plots the year fixed effects. Since higher numerical ratings are worse, positive rising year effects mean stricter standards.

Economic message: This is the core conservatism evidence: the same observable firm is rated more harshly later in time.

Table III—Continued

	Panel A: Ratings Regressions							
	(1) OLS	(2) Ord. Logit	(3) OLS	(4) Ord. Logit	(5) OLS	(6) OLS	(7) OLS	(8) OLS
<i>Idio. Risk</i>			1.806 (-0.01)	2.203 (-0.01)		1.165 (-0.01)	1.199 (-0.01)	1.172 (-0.01)
<i>Linear Trend</i>						0.129 (-0.01)	0.137 (-0.01)	0.137 (-0.01)
<i>Inf. Rate</i>							0.071 (0.11)	
<i>GDP Growth</i>							-0.010 (0.57)	
<i>TS Slope</i>							0.008 (0.24)	
<i>TED Spread</i>							0.011 (0.64)	
<i>P/E</i>							-0.006 (0.36)	
<i>VIX</i>							0.001 (0.90)	
1986	0.334 (-0.01)	0.232 (-0.01)	0.345 (-0.01)	0.357 (-0.01)	0.163 (0.02)	0.228 (-0.01)		
1987	0.565 (-0.01)	0.416 (-0.01)	0.539 (-0.01)	0.538 (-0.01)	0.336 (-0.01)	0.368 (-0.01)	0.437 (-0.01)	
1988	0.614 (-0.01)	0.465 (-0.01)	0.627 (-0.01)	0.662 (-0.01)	0.521 (-0.01)	0.474 (-0.01)		
1989	0.728 (-0.01)	0.557 (-0.01)	0.495 (-0.01)	0.509 (-0.01)	0.597 (-0.01)	0.497 (-0.01)		
1990	0.760 (-0.01)	0.591 (-0.01)	0.462 (-0.01)	0.476 (-0.01)	0.708 (-0.01)	0.619 (-0.01)		
1991	0.705 (-0.01)	0.559 (-0.01)	0.385 (-0.01)	0.392 (-0.01)	0.712 (-0.01)	0.638 (-0.01)		
1992	0.736 (-0.01)	0.540 (-0.01)	0.492 (-0.01)	0.456 (-0.01)	0.694 (-0.01)	0.656 (-0.01)		
1993	0.941 (-0.01)	0.648 (-0.01)	0.693 (-0.01)	0.643 (-0.01)	0.885 (-0.01)	0.730 (-0.01)		
1994	1.047 (-0.01)	0.818 (-0.01)	0.927 (-0.01)	0.855 (-0.01)	0.902 (-0.01)	0.876 (-0.01)		
1995	1.101 (-0.01)	0.875 (-0.01)	1.027 (-0.01)	0.899 (-0.01)	0.900 (-0.01)	0.983 (-0.01)		
1996	1.242 (-0.01)	0.986 (-0.01)	1.150 (-0.01)	1.097 (-0.01)	0.989 (-0.01)	1.084 (-0.01)		
1997	1.378 (-0.01)	1.069 (-0.01)	1.347 (-0.01)	1.255 (-0.01)	1.024 (-0.01)	1.159 (-0.01)		
1998	1.379 (-0.01)	1.045 (-0.01)	1.435 (-0.01)	1.308 (-0.01)	1.062 (-0.01)	1.209 (-0.01)		
1999	1.570 (-0.01)	1.294 (-0.01)	1.601 (-0.01)	1.446 (-0.01)	1.220 (-0.01)	1.347 (-0.01)		
2000	1.938 (-0.01)	1.517 (-0.01)	1.976 (-0.01)	1.801 (-0.01)	1.547 (-0.01)	1.687 (-0.01)		
2001	2.092 (-0.01)	1.621 (-0.01)	2.098 (-0.01)	1.849 (-0.01)	1.788 (-0.01)	1.975 (-0.01)		
2002	2.303 (-0.01)	1.753 (-0.01)	2.194 (-0.01)	1.980 (-0.01)	2.005 (-0.01)	2.060 (-0.01)		
2003	2.674 (-0.01)	2.037 (-0.01)	2.436 (-0.01)	2.230 (-0.01)	2.197 (-0.01)	2.233 (-0.01)		
2004	2.912 (-0.01)	2.342 (-0.01)	2.712 (-0.01)	2.479 (-0.01)	2.429 (-0.01)	2.449 (-0.01)		
2005	3.097 (-0.01)	2.520 (-0.01)	2.878 (-0.01)	2.645 (-0.01)	2.624 (-0.01)	2.699 (-0.01)		
2006	3.241 (-0.01)	2.749 (-0.01)	3.117 (-0.01)	2.892 (-0.01)	2.822 (-0.01)	2.798 (-0.01)		

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Table III, part 1. Ratings regressions show increasing year effects.

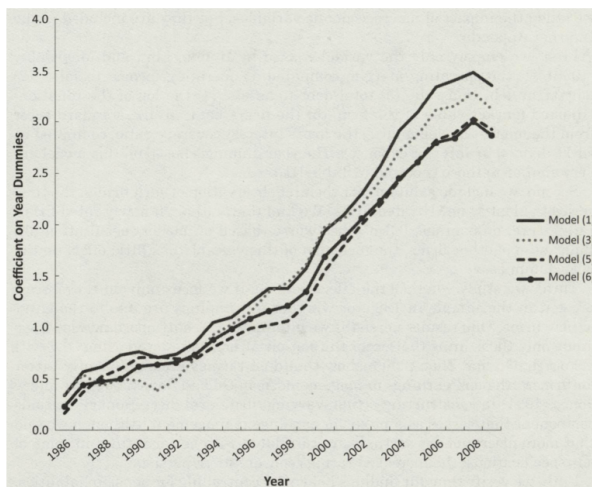


Figure 1. Plot of coefficients on year dummies in ratings model. This figure plots over time the coefficients of the year dummies from models (1), (3), (5), and (6) of Panel A of Table III. These models are estimated using OLS. Models (1) and (3) are estimated with industry fixed effects (three-digit SIC codes) and models (5) and (6) are estimated with firm fixed effects; *Beta* and *Idiosyncratic Risk* are excluded as explanatory variables in models (1) and (5), and included in models (3) and (6).

Figure 1. Year-dummy coefficients rise steadily, visualizing growing conservatism.

Panel A: Ratings Regressions								
	(1) OLS	(2) Ord. Logit	(3) OLS	(4) Ord. Logit	(5) OLS	(6) OLS	(7) OLS	(8) OLS
2007	3.409 (-0.01)	2.811 (-0.01)	3.155 (-0.01)	2.944 (-0.01)	2.889 (-0.01)	2.841 (-0.01)		
2008	3.486 (-0.01)	2.860 (-0.01)	3.288 (-0.01)	3.091 (-0.01)	3.039 (-0.01)	3.002 (-0.01)		
2009	3.354 (-0.01)	2.791 (-0.01)	3.124 (-0.01)	2.967 (-0.01)	2.927 (-0.01)	2.869 (-0.01)		
Industry dummies	Y	Y	Y	Y	N	N	N	N
Firm dummies	N	N	N	N	Y	Y	Y	Y
Observations	28,092	28,092	22,705	22,705	28,092	22,705	22,228	22,228
Number of firms	3,612	3,612	2,906	2,906	3,612	2,906	2,869	2,869
Adj. R ²	0.704	—	0.741	—	0.902	0.903	0.903	0.904
Pseudo R ²	—	0.235	—	0.269	—	—	—	—

Panel B: Marginal Effects of Ordered Logit Model		
Rating	Probability in 1985	Probability in 2009
AAA	0.24%	0.01%
AA+	0.14%	0.01%
AA	0.79%	0.04%
AA-	1.18%	0.06%
A+	3.39%	0.19%
A	9.61%	0.61%
A-	11.54%	0.93%
BBB+	17.42%	2.08%
BBB	22.64%	5.51%
BBB-	14.66%	9.15%
BB+	7.47%	10.98%
BB	6.39%	22.46%
BB-	3.29%	28.33%
B+	1.06%	16.16%
B	0.15%	2.74%
B-	0.03%	0.59%
CCC+	0.01%	0.10%
CCC	0.00%	0.03%
CCC-	0.00%	0.01%
CC	0.00%	0.00%
C	0.00%	0.00%

Table III, part 2. Ordered-logit marginal effects and late-year coefficients.

5.3 Figure 2: Five-Year Default Rates; Table IV: Default and Recovery Regressions

Read: Figure 2 shows category-level default rates; Table IV estimates default and recovery time trends.

Economic message: Default rates decline in many rating categories while recovery rates do not deteriorate. This supports the claim that conservatism is not fully warranted by credit risk.

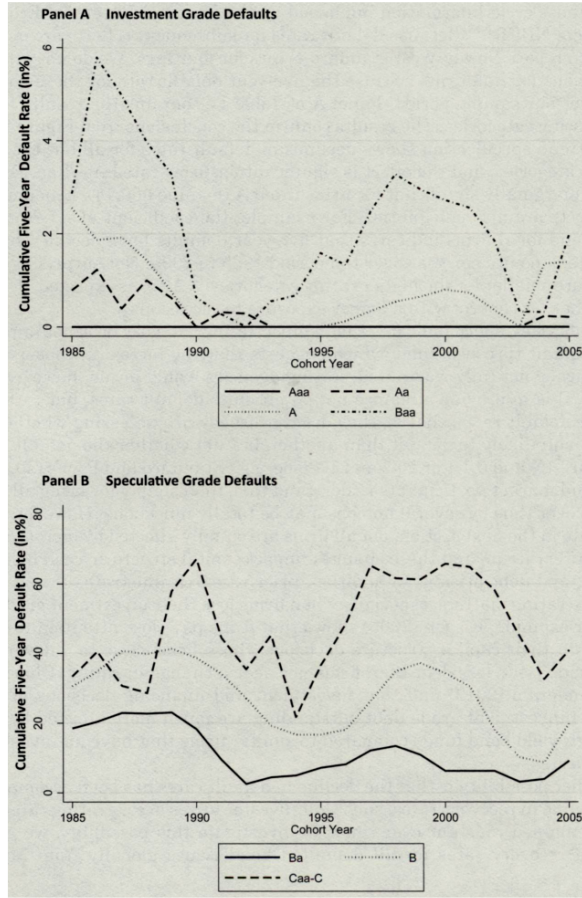


Figure 2. Plot of five-year cumulative default rates by rating category. This figure shows the cumulative five-year issuer-weighted default rates by annual cohort and Moody's rating category from 1985 to 2005. The data are from the Moody's Investors Services report "Corporate Default and Recovery Rates, 1920–2009." *Linear Trend* takes the value of 0 in 1985, 1 in 1986, 2 in 1987, 3 in 1988, etc. *Recession* is the fraction of NBER recession months that a given bond cohort is exposed to over the five-year period. The sample covers bond cohorts from 1985 to 2005. The *p*-values reported below the coefficients are based on Newey-West standard errors estimated with four lags, which we use to account for the overlap in five-year default rates. In Panel B, the dependent variable is the annual issuer-weighted corporate bond recovery rate, based on postdefault trading prices of all corporate bonds. The data are from the Moody's Investors Services report "Corporate Default and Recovery Rates, 1920–2011." *Linear Trend* takes the value of 0 in 1985, 1 in 1986, 2 in 1987, 3 in 1988, etc. The sample covers the years 1985 to 2009. The *p*-values reported in parentheses below the coefficients are based on OLS standard errors.

Table IV
Default and Recovery Rate Regressions

In Panel A, the dependent variable is the cumulative issuer-weighted five-year default rate by annual cohort and Moody's rating category. The data are from the Moody's Investors Services report "Corporate Default and Recovery Rates, 1920–2009." *Linear Trend* takes the value of 0 in 1985, 1 in 1986, 2 in 1987, 3 in 1988, etc. *Recession* is the fraction of NBER recession months that a given bond cohort is exposed to over the five-year period. The sample covers bond cohorts from 1985 to 2005. The *p*-values reported below the coefficients are based on Newey-West standard errors estimated with four lags, which we use to account for the overlap in five-year default rates. In Panel B, the dependent variable is the annual issuer-weighted corporate bond recovery rate, based on postdefault trading prices of all corporate bonds. The data are from the Moody's Investors Services report "Corporate Default and Recovery Rates, 1920–2011." *Linear Trend* takes the value of 0 in 1985, 1 in 1986, 2 in 1987, 3 in 1988, etc. The sample covers the years 1985 to 2009. The *p*-values reported in parentheses below the coefficients are based on OLS standard errors.

Panel A: Multiyear Default Rates by Moody's Rating						
	(1) Aa	(2) A	(3) Baa	(4) Ba	(5) B	(6) Caa-C
Cumulative Five-Year Default Rate (in %)						
<i>Linear Trend</i>	−0.046 (<0.01)	−0.060 (0.14)	−0.141 (0.05)	−0.989 (<0.01)	−1.259 (<0.01)	0.657 (0.23)
<i>Recession</i>	1.508 (<0.01)	5.103 (<0.01)	9.319 (0.03)	45.751 (0.02)	60.708 (0.03)	21.053 (0.69)
<i>Constant</i>	0.567 (<0.01)	0.852 (0.15)	2.692 (<0.01)	16.913 (<0.01)	34.628 (<0.01)	38.425 (<0.01)
Observations	21	21	21	21	21	21
Adjusted R^2	0.508	0.314	0.300	0.750	0.536	0.022
Panel B: Annual Defaulted Corporate Bond Recoveries						
	Recovery Rate (in %)					
<i>Linear Trend</i>	0.091 (0.75)					
<i>Constant</i>	39.872 (<0.01)					
Observations	25					
Adjusted R^2	−0.039					

(2012) and estimate a time-series regression of recovery rates over our sample period against a linear trend variable. As illustrated in Panel B of Table IV, Table IV. Default and recovery rate regressions.

5.4 Table V: Capital Structure Summary; Tables VI–VII: Debt Issuance and Leverage

Read: Table V defines the conservatism variables and policy outcomes. Tables VI and VII show that higher *Rat_Diff* reduces net debt issuance and leverage.

Economic message: Rating conservatism affects financial policy: firms borrow less and maintain lower leverage when ratings are harsher than old standards imply.

Table V
Summary Statistics for Capital Structure and Real Effects Regressions

This table presents annual averages of the variables employed in the capital structure and real effects regressions (variables not reported here are described in Table II). *Rat.Diff.Firm* is the difference between the actual Standard and Poor's rating and the rating predicted by regression model (6) in Panel A of Table III (excluding the year dummies); the credit ratings regression is estimated using data from 1985 to 1996, and the predicted rating is obtained for 1997 to 2009. *Rat.Diff.Ind* is based on ratings regression (3) in Panel A of Table III, and is computed analogously. *Net Debt Issues* are long-term debt issues minus long-term debt reductions, scaled by total assets. *LtdAssets* is long-term debt divided by assets. *Mkt Lev* is total interest-bearing debt divided by the sum of total interest bearing debt and market equity. *Market-to-Book* is the ratio of book value of assets plus market value of equity minus book value of common equity minus balance sheet deferred taxes) to book value of assets. *Carryforwards/Assets* is tax loss carryforwards divided by total assets; missing values of carryforward tax losses are replaced by zero. *R&DSales* is the ratio of R&D expenditures to total sales; missing values of R&D are replaced by zero. *Sales Growth* is the difference in sales between t and $t-1$, divided by sales in year $t-1$. *CAPEX/Sales* is capital expenditures divided by sales. *Acq/Sales* is cash acquisitions divided by sales. *NWC/Assets* is net working capital less cash divided by assets. *Dic. Dummy* equals one in years in which a firm pays a common dividend, zero otherwise. All explanatory variables (except *Rat.Diff.Firm*, *Rat.Diff.Ind*, and *Dic. Dummy*) are winsorized at the 99th percentile; *Net Debt Issues*, *Market-to-Book*, *Sales Growth*, and *NWC/Assets* are also winsorized at the 1st percentile.

Year	<i>Rat.Diff.Firm</i>	<i>Rat.Diff.Ind</i>	<i>Net Debt Issues</i>	<i>LtdAssets</i>	<i>Mkt. Lev</i>	<i>Market-to-Book</i>	<i>Carryforwards/Assets</i>	<i>R&D/Sales</i>	<i>Sales Growth</i>	<i>CAPEX/Sales</i>	<i>Acq/Sales</i>	<i>NWC/Assets</i>	<i>Dic. Dummy</i>
1997	0.183	0.661	0.059	0.361	0.332	1.772	0.029	0.017	0.148	0.171	0.071	0.057	0.513
1998	0.177	0.648	0.067	0.387	0.378	1.708	0.032	0.016	0.141	0.169	0.090	0.048	0.493
1999	0.246	0.847	0.042	0.400	0.380	1.793	0.037	0.015	0.140	0.145	0.067	0.042	0.455
2000	0.525	1.238	0.028	0.375	0.401	1.688	0.041	0.018	0.206	0.147	0.066	0.034	0.442
2001	0.781	1.338	0.025	0.373	0.383	1.590	0.045	0.020	0.060	0.127	0.051	0.033	0.449
2002	0.941	1.500	0.003	0.366	0.391	1.458	0.055	0.021	0.042	0.092	0.034	0.029	0.444
2003	1.101	1.797	0.008	0.358	0.333	1.596	0.061	0.021	0.101	0.087	0.034	0.031	0.450
2004	1.351	2.128	0.013	0.338	0.289	1.669	0.063	0.019	0.161	0.085	0.048	0.035	0.516
2005	1.511	2.289	0.013	0.321	0.280	1.697	0.069	0.019	0.132	0.092	0.053	0.033	0.548
2006	1.720	2.480	0.024	0.321	0.275	1.725	0.075	0.019	0.128	0.105	0.062	0.037	0.570
2007	1.728	2.493	0.034	0.325	0.293	1.737	0.067	0.019	0.121	0.116	0.072	0.031	0.573
2008	1.961	2.952	0.025	0.340	0.419	1.327	0.073	0.020	0.112	0.112	0.047	0.027	0.560
2009	1.825	2.429	-0.006	0.335	0.355	1.436	0.080	0.020	-0.090	0.103	0.021	0.030	0.530
Mean	0.937	1.695	0.028	0.355	0.347	1.634	0.055	0.019	0.109	0.119	0.055	0.036	0.500
N	7,274	14,238	16,787	18,146	18,890	14,659	18,152	18,122	16,188	17,922	16,772	17,428	17,959

Table V. Conservatism measures and capital-structure summary statistics.

Table VI
Capital Structure Regressions: Leverage Changes

This table reports the coefficients for regression models of leverage changes. L(.) denotes the lag operator. Standard errors are clustered at the firm level and adjusted for heteroskedasticity and autocorrelation. *p*-values are reported in parentheses below the coefficients. *Taxshield* is computed as investment tax credits divided by assets. The other variables are defined in Tables II and V.

	<i>Net Debt Issues</i>			
	(1)	(2)	(3)	(4)
<i>L2.(Rat.Diff.Ind)</i>	-0.002 (<0.01)			
<i>L2.(Rat.Diff.Firm)</i>		-0.003 (<0.01)		
<i>L2.(Rat.Diff.Ind) - L3.(Rat.Diff.Ind)</i>			-0.002 (0.05)	
<i>L3.(Rat.Diff.Ind)</i>			-0.002 (0.03)	
<i>L2.(Rat.Diff.Firm) - L3.(Rat.Diff.Firm)</i>				-0.003 (0.03)
<i>L3.(Rat.Diff.Firm)</i>				-0.002 (<0.01)
<i>L. Rating</i>	0.002 (0.01)	0.001 (0.07)	0.002 (0.07)	0.001 (0.20)
<i>L. Book Lev</i>	-0.073 (<0.01)	-0.066 (<0.01)	-0.065 (<0.01)	-0.064 (<0.01)
<i>L.(Market-to-Book)</i>	0.013 (<0.01)	0.011 (<0.01)	0.013 (<0.01)	0.011 (<0.01)
<i>L.(PPE/Assets)</i>	0.014 (0.09)	0.004 (0.71)	0.015 (0.08)	0.001 (0.95)
<i>L.Profit</i>	0.030 (0.02)	0.031 (0.08)	0.043 (<0.01)	0.036 (0.05)
<i>L.Size</i>	-0.003 (0.02)	-0.005 (<0.01)	-0.003 (0.04)	-0.005 (<0.01)
<i>L.Taxshield</i>	-5.842 (<0.01)	-4.634 (<0.01)	-4.368 (0.01)	-3.103 (0.10)
<i>L.(Carryforwards/Assets)</i>	-0.013 (0.19)	-0.002 (0.87)	-0.017 (0.07)	-0.0002 (0.99)
<i>L.(R&D/Sales)</i>	-0.042 (0.27)	-0.046 (0.44)	-0.062 (0.12)	-0.070 (0.21)
Industry dummies	Y	Y	Y	Y
Year dummies	Y	Y	Y	Y
Observations	9,769	6,001	8,606	5,635
Number of firms	1,754	887	1,554	829
Adjusted R^2	0.070	0.072	0.065	0.066

These findings indicate that the increased ratings stringency has had a substantial impact on firms' capital structure decisions. They may also add to our understanding of why a considerable number of firms seem to be underlevered. Table VI. Conservatism reduces net debt issues.

Table VII
Capital Structure Regressions: Leverage Levels

This table reports the coefficients for regression models of leverage levels. $L(.)$ denotes the lag operator. Standard errors are clustered at the firm level and adjusted for heteroskedasticity and autocorrelation. p -values are reported in parentheses below the coefficients. $Taxshield$ is computed as investment tax credits divided by assets. The other variables are defined in Tables II and V.

	(1)	(2)	(3)	(4)	(5)	(6)
	<i>Ltde/Assets</i>	<i>Book_Lev</i>	<i>Mkt_Lev</i>	<i>Ltde/Assets</i>	<i>Book_Lev</i>	<i>Mkt_Lev</i>
<i>L.(Rat.Diff_Ind)</i>	-0.052 (<0.01)	-0.059 (<0.01)	-0.040 (<0.01)			
<i>L.(Rat.Diff_Firm)</i>				-0.012 (<0.01)	-0.011 (<0.01)	-0.005 (0.03)
<i>Rating</i>	0.053 (<0.01)	0.059 (<0.01)	0.057 (<0.01)	0.030 (<0.01)	0.032 (<0.01)	0.039 (<0.01)
<i>Market-to-Book</i>	0.014 (<0.01)	0.016 (<0.01)	-0.093 (<0.01)	0.005 (0.47)	0.008 (0.28)	-0.083 (<0.01)
<i>PPE/Assets</i>	0.137 (<0.01)	0.142 (<0.01)	0.101 (<0.01)	0.026 (0.43)	0.018 (0.61)	-0.003 (0.92)
<i>Profit</i>	0.129 (<0.01)	0.141 (<0.01)	0.055 (<0.01)	0.001 (0.98)	0.005 (0.92)	-0.074 (0.03)
<i>Size</i>	0.034 (<0.01)	0.050 (<0.01)	0.059 (<0.01)	-0.002 (0.59)	0.008 (0.11)	0.032 (<0.01)
<i>Taxshield</i>	-0.894 (0.88)	1.974 (0.73)	7.617 (0.03)	-2.205 (0.77)	1.041 (0.89)	8.532 (0.07)
<i>Carryforwards/Assets</i>	0.047 (0.08)	0.064 (0.01)	0.070 (<0.01)	0.072 (0.23)	0.108 (0.06)	0.110 (<0.01)
<i>R&D/Sales</i>	-0.290 (<0.01)	-0.305 (<0.01)	-0.351 (<0.01)	-0.201 (0.21)	-0.209 (0.22)	-0.299 (0.02)
Industry dummies	Y	Y	Y	Y	Y	Y
Year dummies	Y	Y	Y	Y	Y	Y
Observations	11,832	11,831	11,831	6,730	6,729	6,729
Number of firms	1,999	1,999	1,999	935	935	935
Adjusted R^2	0.588	0.627	0.696	0.544	0.529	0.672

ered despite the tax benefits of debt (see, e.g., Graham (2000)). Of course, our findings can neither explain the zero leverage puzzle (see, e.g., Strebulaev and Yang (2013)), nor can they explain why firms had low leverage at the start of our sample period when rating agencies were not as strict compared to later years.²¹

Next, we examine whether the increase in conservatism has affected firms' decisions to access the bond market altogether. We borrow Faulkender and Petersen's (2006) model of bond market access and, using all firms in the Compustat database with sufficient information available to compute the explanatory variables, we estimate a logit model of the likelihood of having a rating

²¹ The effect of conservatism on debt issuance and leverage is also more pronounced for small firms than for large firms (reported in the Internet Appendix), consistent with the view that ratings

Table VII. Conservatism lowers book leverage, long-term debt/assets, and market leverage.

5.5 Tables VIII–IX: Bond Market Access and Cash Holdings

Read: Table VIII shows reduced bond-rating access, especially for firms affected by conservatism. Table IX shows higher cash/assets for firms facing harsher ratings.

Economic message: Firms respond to rating conservatism on two margins: they are less likely to use public debt markets and they self-insure with cash.

Table VIII
Determinants of Bond Market Access

This table reports the coefficients for logit regression models of bond market access. The dependent variable is a dummy variable set equal to one if the firm has a Standard and Poor's long-term issuer rating in a given year. *S&P500* is a dummy for S&P500 index firms. *NYSE* is a dummy for firms traded on the NYSE. *Ln(MVA)* is the log of market value of assets (computed as book value of assets – book value of equity + market value of equity). *Ln(Age + 1)* is the log of one plus the number of years the firm has been included in the Compustat database. *Ln(%Rated + 1)* is the log of one plus the percentage of firms in the same three-digit SIC industry that have a bond rating. *Young Firm* indicates whether the firm is three years old or less (0/1 dummy). *Market-to-Book* is market value of assets divided by book value assets. *Advertising/Sales* is advertising expenses divided by sales. *Asset.Vol* is annualized volatility of monthly equity returns over the past year, multiplied by (market value equity/market value of assets). *Ann. Ret.* is the annual stock return over the previous year. *Rat.Diff.Predictions* is the firm's predicted rating using regression model (1) in Panel A of Table III, excluding year dummies and estimated over the period 1997 to 2009, minus the predicted rating using the same model estimated over the period 1985 to 1996. The other variables are defined in Tables II and V. The sample spans 1985 to 2009 in column (1), and 1997 to 2009 in column (2). Standard errors are clustered at the firm level and adjusted for heteroskedasticity and autocorrelation. *p*-values are reported in parentheses next to the coefficients.

	Rating Dummy	
	(1)	(2)
<i>S&P500</i>	0.769 (<0.01)	1.372 (<0.01)
<i>NYSE</i>	0.366 (<0.01)	0.267 (<0.01)
<i>Ln(MVA)</i>	0.976 (<0.01)	1.065 (<0.01)
<i>Ln(Age + 1)</i>	0.218 (<0.01)	0.307 (<0.01)
<i>Ln(%Rated + 1)</i>	1.435 (<0.01)	1.211 (<0.01)
<i>Young Firm</i>	-0.260 (<0.01)	0.010 (0.93)
<i>Profit</i>	-0.194 (<0.01)	0.526 (<0.01)
<i>PPE/Assets</i>	-0.675 (<0.01)	-0.025 (0.91)
<i>Market-to-Book</i>	-0.425 (<0.01)	-0.434 (<0.01)
<i>Advertising/Sales</i>	0.088 (0.94)	-1.741 (0.34)
<i>Asset.Vol</i>	-1.838 (<0.01)	0.431 (<0.01)
<i>R&D/Sales</i>	-0.378 (0.05)	-0.777 (<0.01)
<i>Ann. Ret.</i>	0.091 (<0.01)	0.054 (0.06)
1986	0.428 (<0.01)	
1987	0.457 (<0.01)	
1988	0.206 (0.03)	
1989	-0.077 (0.44)	
1990	-0.174 (0.08)	
1991	-0.171 (0.09)	
1992	-0.169 (0.10)	
1993	-0.107 (0.28)	
1994	-0.179 (0.08)	
1995	-0.151 (0.14)	
1996	-0.194 (0.06)	
1997	-0.206 (0.05)	
1998	-0.202 (0.07)	

Table VIII, part 1. Bond-market access regressions.

Table IX
Cash Regressions

This table reports the coefficients for regression models of cash holdings. *L(.)* denotes the lag operator. Standard errors are clustered at the firm level and adjusted for heteroskedasticity and autocorrelation. *p*-values are reported in parentheses below the coefficients. *Taxshield* is computed as investment tax credits divided by assets. The other variables are defined in Tables II and V.

	Cash / Assets	
	(1)	(2)
<i>L.(Rat.Diff.Ind)</i>	0.006 (<0.01)	
<i>L.(Rat.Diff.Firm)</i>		0.002 (0.04)
<i>Rating</i>	-0.004 (<0.01)	-0.001 (0.14)
<i>Acq./Sales</i>	-0.068 (<0.01)	-0.053 (<0.01)
<i>Div. Dummy</i>	-0.002 (0.54)	-0.006 (0.19)
<i>CAPEX/Sales</i>	0.034 (<0.01)	0.001 (0.93)
<i>NWC/Assets</i>	-0.172 (<0.01)	-0.173 (<0.01)
<i>Vol</i>	0.098 (<0.01)	0.115 (<0.01)
<i>Market-to-Book</i>	0.023 (<0.01)	0.016 (<0.01)

Table IX. Conservatism is associated with higher cash holdings.

Table VIII—Continued

	Rating Dummy	
	(1)	(2)
2003	-0.792 (<0.01)	
2004	-1.021 (<0.01)	
2005	-1.117 (<0.01)	
2006	-1.217 (<0.01)	
2007	-1.360 (<0.01)	
2008	-1.224 (<0.01)	
2009	-1.234 (<0.01)	
<i>Rat.Diff.Predictions</i>		-1.141 (<0.01)
Industry dummies	Y	Y
Observations	90,726	43,168
Number of firms	12,212	7,877
Pseudo R^2	0.507	0.554

by one notch, the likelihood of having a bond rating declines by nine percentage points.

Finally, we investigate whether conservatism has affected cash holdings.

Table VIII, part 2. Continuation with access coefficients.

5.6 Tables XI–XII: Credit Spread Evidence

Read: Table XI summarizes the bond-spread sample; Table XII estimates spreads with actual ratings and the conservatism measure.

Economic message: Conditional on actual rating, firms rated worse than old standards predict have lower spreads. Investors partly see through rating conservatism.

Table XII
Credit Spread Regressions with Ratings Difference Based on Firm Dummies

This table reports the coefficients for panel regression models of credit spreads (dependent variable in all specifications: *Yield Spread*). The explanatory variables are defined in Table XI. *p*-values (based on standard errors clustered by bond and adjusted for heteroskedasticity and autocorrelation) are reported in parentheses below the coefficients.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
<i>Rat.Diff.Firm</i>	-0.095 (-0.01)	-0.201 (-0.01)	-0.115 (-0.01)	-0.198 (-0.01)	-0.148 (-0.01)	-0.074 (-0.01)	-0.187 (-0.01)	-0.116 (-0.01)
<i>Issue Rating</i>	0.741 (-0.01)	0.848 (-0.01)	0.750 (-0.01)	0.846 (-0.01)	0.559 (-0.01)	0.507 (-0.01)	0.593 (-0.01)	0.544 (-0.01)
<i>Ln(Days to Maturity)</i>			0.626 (-0.01)	0.646 (-0.01)	0.630 (-0.01)	0.722 (-0.01)	0.630 (-0.01)	0.726 (-0.01)
<i>Equity.Vol</i>					69.551 (-0.01)	71.933 (-0.01)	68.910 (-0.01)	70.515 (-0.01)
<i>EDF</i>					0.055 (-0.01)	0.055 (-0.01)	0.056 (-0.01)	0.055 (-0.01)
<i>Ann. Ret.</i>					-0.004 (-0.01)	-0.003 (-0.01)	-0.004 (-0.01)	-0.003 (-0.01)
<i>Profit</i>							-0.446 (-0.01)	-0.633 (-0.01)
<i>Book.Lev</i>							-1.753 (-0.01)	-1.420 (-0.01)
<i>PPE / Assets</i>							1.554 (-0.01)	1.430 (-0.01)
Firm dummies	Y	N	Y	N	Y	N	Y	N
Bond dummies	N	Y	N	Y	N	Y	N	Y
Year-Month dummies	Y	Y	Y	Y	Y	Y	Y	Y
Observations	197,662	197,662	197,662	197,662	195,995	195,995	193,625	193,625
Number of firms	701	701	701	701	698	698	694	694
Number of bonds	4,864	4,864	4,864	4,864	4,836	4,836	4,795	4,795
Adjusted R²	0.637	0.728	0.663	0.730	0.775	0.816	0.776	0.816

its covenants, and imbedded options²⁸; we also include monthly time dummies to control for any macroeconomic factors. All standard errors are clustered at the bond level and adjusted for heteroskedasticity and autocorrelation.

Table XII contains the results. As expected, debt spreads increase as ratings worsen. In model (1), which only includes the bond's actual rating and the difference between the actual and predicted ratings, spreads increase by Table XII. Spread regressions show partial market reversal of conservative ratings.

Table XI
Summary Statistics for Credit Spread Regressions

This table presents annual averages of the variables employed in the credit spread regressions. The sample consists of 701 unique issuers and 4,864 unique bonds over the sample period January 1997 to June 2009. *Issue Rating* is the issue-specific rating from Standard and Poor's. *Rat.Diff.Firm* is the difference between the actual Standard and Poor's rating and the rating predicted by regression model (6) in Panel A of Table III (excluding year dummies); the credit ratings regression is estimated using data from 1985 to 1996, and the predicted rating is obtained for 1997 to 2009. *Bond Yield* is the yield-to-maturity on the corporate bond (based on Merrill Lynch calculations). *Treasury Yield* is the yield on the five-year U.S. Treasury bond. *Yield Spread* is the difference between *Bond Yield* and *Treasury Yield*. *Ln(Days to Maturity)* is the natural logarithm of the number of days to maturity of a given bond. *Equity.Vol* is the standard deviation of daily stock returns computed over the past 12 months. *EDF* is the expected default frequency estimated following Bharath and Shumway (2008). *Ann. Ret.* is the firm's stock return over the previous year, calculated using CRSP monthly data. *Profit* is operating income before depreciation divided by sales (from quarterly Compustat). *Book.Lev* is the sum of debt in current liabilities and long-term debt divided by assets (from quarterly Compustat). *PPE/Assets* is net property, plant, and equipment divided by assets (from quarterly Compustat). *Bond Yield*, *Equity.Vol*, *Ann. Ret.*, *Profit*, *Book.Lev*, and *PPE/Assets* are winsorized at the 99th percentile, while *Equity.Vol*, *Ann. Ret.*, and *Profit* are additionally winsorized at the 1st percentile.

Year	Issue Rating	Rat.Diff.Firm	Bond Yield	Treasury Yield	Yield Spread	Ln(Days to Maturity)	Equity.Vol	EDF	Ann. Ret.	Profit	Book.Lev	PPE/Assets
1997	8.260	0.500	7.351	6.123	1.228	8.107	0.019	1.705	25.055	0.187	0.350	0.445
1998	8.119	0.119	6.869	5.074	1.796	8.103	0.021	2.103	20.203	0.190	0.361	0.426
1999	8.177	0.558	7.646	5.559	2.087	8.098	0.027	4.323	9.169	0.185	0.368	0.411
2000	8.532	0.666	8.956	6.084	2.872	8.054	0.030	8.457	-1.885	0.188	0.381	0.390
2001	8.723	0.857	7.990	4.432	3.558	7.982	0.030	8.373	10.219	0.178	0.376	0.391
2002	8.974	1.160	7.304	3.659	3.645	7.946	0.027	8.247	1.349	0.167	0.376	0.396
2003	9.170	1.340	5.942	2.922	3.019	7.910	0.027	8.378	-1.007	0.172	0.374	0.389
2004	9.340	1.624	5.645	3.419	2.226	7.862	0.018	9.603	34.046	0.180	0.360	0.383
2005	9.291	1.959	5.856	4.047	1.809	7.811	0.016	3.276	17.518	0.191	0.350	0.364
2006	9.306	2.022	6.450	4.732	1.718	7.838	0.016	2.153	16.180	0.198	0.340	0.356
2007	9.236	2.026	6.403	4.847	1.557	7.839	0.016	0.702	20.938	0.195	0.319	0.356
2008	8.997	1.955	7.919	2.753	5.166	7.895	0.025	8.122	-11.861	0.185	0.320	0.356
2009	9.200	2.054	8.729	2.081	6.648	7.845	0.043	22.557	-40.830	0.179	0.333	0.376
Mean	8.841	1.258	7.072	4.304	2.768	7.965	0.024	6.561	10.022	0.184	0.357	0.389
N	197,838	197,838	197,662	197,838	197,662	197,837	197,838	196,171	196,791	194,703	196,987	197,080

Table XI. Credit-spread sample summary statistics.

5.7 Table XIII: Robustness Tests - Alternative Conservatism Measure

Read: Table XIII redefines conservatism using old versus new model predictions and repeats the main corporate-policy and spread tests.

Economic message: The results survive alternative measurement of conservatism, strengthening the paper's interpretation.

Table XIII
Robustness Tests: Alternative Measure of Conservatism

This table reports robustness tests for the capital structure and credit spread regression using an alternative measure of conservatism. *Rat.Diff.New.Firm* is the difference between the actual Standard and Poor's rating and the rating predicted by regression model (6) in Panel A of Table III (excluding year dummies); the credit ratings regression is estimated using data from 1997 to 2009, and the predicted rating is obtained for 1997 to 2009. *Rat.Diff.New.Firm* is the difference between the predicted rating based on estimating the ratings model over 1997 to 2009 and the predicted rating based on estimating the ratings model over 1985 to 1996; the ratings model employed is based on regression (6) in Panel A of Table III. Panel A reports regression models of leverage changes. The control variables are the same as in Table VI. Panel B reports regression models of leverage levels. The control variables are the same as in Table VII. Standard errors in Panels A and B are clustered at the firm level and adjusted for heteroskedasticity and autocorrelation. Panel C reports regression models of credit spreads. The control variables are the same as in Table XII. Standard errors are clustered at the bond level and adjusted for heteroskedasticity and autocorrelation. *L(.)* denotes the lags operator in all panels. *p*-values in all panels are reported in parentheses below the coefficients.

Panel A: Debt Issuance: Robustness	
	Net Debt Issues
<i>L2.(Rat.Diff.Predictions.Firm)</i>	-0.002 (0.02)
<i>L2.(Rat.Diff.New.Firm)</i>	-0.005 (-0.01)
<i>L.Rating</i>	0.002 (0.02)
<i>L.Book.Lev</i>	-0.067 (-0.01)
<i>L.(Market-to-Book)</i>	0.011 (-0.01)
<i>L.(PPE/Assets)</i>	0.004 (0.70)
<i>L.Profit</i>	0.003 (0.06)
<i>L.Size</i>	-0.005 (-0.01)
<i>L.Taxshield</i>	-4.574 (0.01)
<i>L.(Carryforwards/Assets)</i>	-0.004 (0.77)
<i>L.(R&D/Sales)</i>	-0.047 (0.43)
Industry and Year dummies	Y
Observations	5,994
Number of firms	881
Adjusted R ²	0.073

(Continued)

Table XIII—Continued

Panel B: Debt Levels: Robustness

	(1) <i>Ltd/Assets</i>	(2) <i>Book.Lev</i>	(3) <i>Mkt.Lev</i>
<i>L.(Rat.Diff.Predictions.Firm)</i>	-0.011 (-0.01)	-0.009 (-0.01)	-0.002 (0.53)
<i>L.(Rat.Diff.New.Firm)</i>	-0.014 (-0.01)	-0.017 (-0.01)	-0.012 (-0.01)
<i>Rating</i>	0.000 (-0.01)	0.002 (-0.01)	0.040 (-0.01)
<i>Market-to-Book</i>	0.005 (0.44)	0.009 (0.23)	-0.062 (-0.01)
<i>PPE/Assets</i>	0.027 (0.43)	0.019 (0.59)	-0.001 (0.97)
<i>Profit</i>	0.003 (0.95)	0.009 (0.85)	-0.067 (0.05)
<i>Size</i>	-0.003 (0.55)	0.008 (0.12)	0.002 (-0.01)
<i>Taxshield</i>	-1.879 (0.80)	1.691 (0.83)	9.459 (0.05)
<i>Carryforwards/Assets</i>	0.071 (0.24)	0.105 (0.07)	0.104 (-0.01)
<i>R&D/Sales</i>	-0.202 (0.21)	-0.202 (0.21)	-0.302 (0.02)
Industry and Year dummies	Y	Y	Y
Observations	6,725	6,724	6,724
Number of firms	930	930	930
Adjusted R ²	0.544	0.530	0.673

Panel C: Credit Spreads: Robustness

	(1) <i>Yield Spread</i>	(2) <i>Yield Spread</i>
<i>Rat.Diff.Predictions.Firm</i>	-0.278 (-0.01)	-0.288 (-0.01)
<i>Rat.Diff.New.Firm</i>	-0.183 (-0.01)	-0.111 (-0.01)
<i>Issue Rating</i>	0.569 (-0.01)	0.565 (-0.01)
<i>Ln(Days to Maturity)</i>	0.630 (-0.01)	0.725 (-0.01)
<i>Equity.Vol</i>	70.002 (-0.01)	71.914 (-0.01)

Table XIII, part 2. Leverage robustness.

Table XIII—Continued

Panel C: Credit Spreads: Robustness

	(1) <i>Yield Spread</i>	(2) <i>Yield Spread</i>
<i>Book.Lev</i>	-1.868 (-0.01)	-1.588 (-0.01)
<i>PPE/Assets</i>	1.455 (-0.01)	1.261 (-0.01)
Firm dummies	Y	N
Bond dummies	N	Y
Year-Month dummies	Y	Y
Observations	193,492	193,492
Number of firms	679	679
Number of bonds	4,772	4,772
Adjusted R ²	0.777	0.817

current model, spreads also decline by 11.1 basis points. This compares to an increase in spreads of 54.5 basis points per rating notch decline. According to this specification, the market undoes 49% (26.8/54.5) of the effect of ratings conservatism on debt spreads. In sum, using an alternative measure of ratings conservatism based on the

Table XIII, part 3. Spread robustness.

6 Short summary

- Corporate issuer ratings became much more conservative from 1985 to 2009.
- The trend survives extensive controls and firm fixed effects.
- Default rates within rating categories decline, implying the tightening is not fully warranted.
- Firms most affected by conservatism issue less debt, have lower leverage, hold more cash, and access bond markets less.
- Bond investors partially correct conservative ratings through lower spreads for firms rated worse than old standards predict.
- Ratings therefore affect capital structure and debt pricing, but capital markets do not treat them mechanically.

7 Conclusion

7.1 Core conclusion

Baghai, Servaes, and Tamayo show that corporate rating agencies became more conservative, not more lenient, over 1985-2009. The same observable firm received progressively worse ratings, and this tightening appears only partly warranted by risk.

7.2 Why it matters

Ratings influence debt policy. A harsher rating regime lowers debt issuance, leverage, public bond access, and raises cash holdings. This means rating standards help shape firms' capital structures.

7.3 Debt-pricing interpretation

Bond markets partly undo conservative ratings: conditional on actual ratings, firms with ratings worse than old-standard predictions have lower spreads. But the correction is incomplete, which explains why firms still adjust real policies.

7.4 Incremental takeaway

The paper changes the post-crisis ratings narrative. Rating agencies may inflate ratings in one market, such as structured finance, while becoming conservative in another, such as corporate debt. Rating standards are market-specific, time-varying inputs into corporate finance.

7.5 Limitations

The paper cannot fully explain why conservatism rose. It also cannot observe every risk characteristic. Its strongest evidence comes from the convergence of controlled rating regressions, default trends, firm policy effects, and bond spread pricing.

7.6 Final takeaway

Credit ratings are not passive summaries of fundamentals. They are institutional constraints that can alter financing choices and asset prices when rating standards shift.